

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF COMPLIANCE****Note:** This table completed by **HERS****ECC** Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. General Information**Note:** If Kitchen Range Hood installation (new or replacement) is selected in A13, **HERS****ECC** verification and a CF2R/3R-MCH-32 is required.

01	Project Name:		02	Date Prepared:	
03	Project Location:		04	Building Front Orientation (deg or cardinal):	
05	CA City:		06	Number of Altered Dwelling Units:	
07	Zip Code:		08	Fuel Type:	
09	Climate Zone:		10	Total Conditioned Floor Area (ft ²):	
11	Building Type:		12	Slab Area (ft ²):	
13	Project Scope:				

B. Building Insulation-Opaque Surface Details – Framed (Section 150.2(b)1)**Note:**

Where insulation is installed above the roofing membrane, or above the layer used to seal the roof from water penetration, the insulation shall have a maximum water absorption of 0.3 percent by volume when tested according to American Society for Testing and Materials (ASTM) Standard C272.

01	02	03	04	05	06a	06b	07	08	09	1008	1109
Tag/ID	Assembly Type	Frame Type	Frame Depth (inches)	Frame Spacing (inches)	Proposed					Required	Comments
					Cavity R-value	Continuous Insulation R-value	U-factor	Joint Appendix JA4 Reference		U-Factor or R-value	
								Table	Cell		

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****C. Opaque Surface ~~Building Insulation~~ Details – Nonframed**

01	02	03	04	05	06	07	10 08	11 09
Tag/ID	Assembly Type	Assembly Materials	Thickness (inches)	Proposed			Required	Comments
				Core Insulation R-value	Continuous Insulation R-value	U-factor	Required	
							U-factor or R-value	

D. Opaque Surface ~~Building Insulation~~ Details – Masonry/Mass Walls

Note:

When insulation is added to the outside of a mass wall and/or when the inside is furred and insulated, the performance data may be adjusted using Equation 4-4 in the Joint Appendices.

01	02	03	04	05	06	07	08	09	10	11	12	13	14	15	16
Tag/ID	Above or Below Grade?	<u>Masonry</u> /Mass Type	<u>Masonry</u> /Mass Thickness (inches)	Proposed								Required			
				<u>Mass Wall</u> Joint Appendix JA4 Reference		Exterior Insulation		Interior Insulation		<u>Insulation Layer</u> Joint Appendix JA4 Reference		Exterior Insulation		Interior Insulation	
				Table	Cell	R-value	U-factor	R-value	U-factor	Table	Cell	R-value	U-factor	R-value	U-factor

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****E. Roof Replacement (Section 150.2(b)1H1I)**

Note:

- Roof area covered by building integrated photovoltaic panels and solar thermal panels are exempt from the above Cool Roof requirements.
- Liquid field applied coatings must comply with installation criteria from Section 110.8(i)4.

01	02	03	04	05	06	07	08	09	10	11	12	13	14
Tag/ ID	Method of Compliance	Roof Pitch	Exception	CRRC Product ID Number	Product Type	R-value Deck Insulation	Proposed				Minimum Required		
							Initial Solar Reflectance	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)

F. Fenestration/Glazing Allowed Areas and Efficiencies (Section 150.2(b)1A)

01	02	03	04		05		06		07
Alteration Type	Maximum Allowed Fenestration Area for All Orientations (ft ²)	Maximum Allowed West-Facing Fenestration Area Only (ft ²)	Existing Fenestration Area for All Orientations (ft ²)	Existing West-Facing Fenestration Area (ft ²)	Maximum Allowed U-factor (Windows)	Maximum Allowed U-factor (Skylights)	Maximum Allowed SHGC (Windows)	Maximum Allowed SHGC (Skylights)	Comments

G. Fenestration/Glazing Proposed Areas and Efficiencies – Add (Section 150.2(b)1A)

Note: Doors with greater than or equal to 25 percent glazed area are considered glazed doors and are treated as fenestration products.

01	02	03	04	05	06	07	08	09	10	11	12	13	14
Tag/ID	Fenestration Type	Frame Type	Dynamic Glazing	Orientation N, S, W, E	Number of Panes	Proposed Fenestration Area (ft ²)	Proposed West Facing Fenestration Area (ft ²)	Proposed U-factor	Proposed U-factor Source	Proposed SHGC	Proposed SHGC Source	Exterior Shading Device	Combined SHGC from CF1R-ENV-03
15	Existing + Proposed Fenestration Area												

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16	Maximum Allowed Fenestration Area		
17	Compliance Statement:		
18	Existing + Proposed West-Facing Fenestration Area		
19	Maximum Allowed West Fenestration Area		
20	Compliance Statement:		
21	Proposed Fenestration U-factor (Windows)		
22	Required Fenestration U-factor (Windows)		
23	Compliance Statement:		
24	Proposed Fenestration SHGC (Windows)		
25	Required Fenestration SHGC (Windows)		
26	Compliance Statement:		
27	Proposed Fenestration U-factor (Skylights)		
28	Required Fenestration U-factor (Skylights)		
29	Compliance Statement:		
30	Proposed Fenestration SHGC (Skylights)		
31	Required Fenestration SHGC (Skylights)		
32	Compliance Statement:		

H. Fenestration/Glazing Proposed Areas and Efficiencies – Replace (Section 150.2(b)1B)

Note: Doors with greater than or equal to 25 percent glazed area are considered glazed doors and are treated as fenestration products.

01	02	03	04	05	06	07	08	9	10	11	12	13	14
Tag/ ID	Fenestration Type	Frame Type	Dynamic Glazing	Orientation N, S, W, E	Area Removed (ft ²)	Area Added (ft ²)	Net Added Area (ft ²)	Proposed U-factor	Proposed U-factor Source	Proposed SHGC	Proposed SHGC Source	Exterior Shading Device	Combined SHGC from CF1R-ENV-03
15	Net Added West-facing Fenestration Area												
16	Is Net Added Fenestration Area ≤ 0 for West-Facing Fenestration?												
17	Net Added Fenestration Area (all orientations)												

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18	Is Net Added Fenestration Area ≤ 0 for All Orientations?	
19	Proposed Fenestration U-factor (Windows)	
20	Required Fenestration U-factor (Windows)	
21	Compliance Statement:	
22	Proposed Fenestration SHGC (Windows)	
23	Required Fenestration SHGC (Windows)	
24	Compliance Statement:	
25	Proposed Fenestration U-factor (Skylights)	
26	Required Fenestration U-factor (Skylights)	
27	Compliance Statement:	
28	Proposed Fenestration SHGC (Skylights)	
29	Required Fenestration U-factor (Skylights)	
30	Compliance Statement:	

H1. Opaque Swinging Doors to Exterior (Section 150.1(c)5)

Notes:

- Any door with 25 percent or more glass is counted as a fenestration product in Tables F,G and H.
- Do not include fire-rated doors between garage or unconditioned space and conditioned space.
- If using weighted average to achieve required maximum U-factor, attach CF1R-ENV-02-E.

01	02	03	04	05	06	07
Tag/ID	Area	Proposed U-factor	Proposed U-factor Source	Required Maximum U-factor	Weighted Average (Yes/No)	Comments

H2. Space Conditioning (SC) Systems - Heating/Cooling (Section 150.2(b))

01	02	03
Dwelling Unit Name	Dwelling Unit Total CFA (ft ²)	Comments

Registration Number:

Registration Date/Time:

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****JK. Water Heating Systems** (Section 150.2(b)1H)

List water heaters and boilers for both domestic hot water (DHW) heaters and hydronic space heating.

Options:

1. Gas or propane water heating system; or
2. A single heat pump water heater. The storage tank shall not be located outdoors and shall be placed on an incompressible, rigid insulated surface with a minimum thermal resistance of R-10. The water heater shall be installed with a communication interface that meets either the requirements of Section 110.12(a) or has a ANSI/CTA-2045-B communication port; or
3. A single heat pump water heater that meets the requirements of NEEA Advanced Water Heater Specification Tier 3 or higher; or
4. If no natural gas is connected to the existing water heater location, a consumer electric water heater (Heat Pump Water Heater or Electric Resistance).

01	<u>Is natural gas connected to the existing water heater?</u> Is the existing water heater electric resistance?			
02	04 <u>03</u>	05 <u>04</u>	07 <u>05</u>	08 <u>06</u>
Water Heating System ID or Name	System Option (from §150.2(b)1Hiii)	Water Heater Type	Fuel Type	# of Water Heaters in System

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**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****DOCUMENTATION AUTHOR'S DECLARATION STATEMENT**

1. I certify that this Certificate of Compliance documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Company:	Signature Date:
Address:	CEA/AEA/ HERS ECC Certification Identification (if applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Compliance is true and correct.
2. I am eligible under Division 3 of the Business and Professions Code to accept responsibility for the building design or system design identified on this Certificate of Compliance (responsible designer).
3. The energy features and performance specifications, materials, components, and manufactured devices for the building design or system design identified on this Certificate of Compliance conform to the requirements of Title 24, Part 1 and Part 6 of the California Code of Regulations.
4. The building design features or system design features identified on this Certificate of Compliance are consistent with the information provided on other applicable compliance documents, worksheets, calculations, plans and specifications submitted to the enforcement agency for approval with this building permit application.
5. I understand that a registered copy of this Certificate of Compliance shall be made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to accomplish this requirement.
6. I understand that a registered copy of this Certificate of Compliance is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to accomplish these requirements.

Responsible Designer Name:	Responsible Designer Signature:
Company:	Date Signed:
Address:	License:
City/State/Zip:	Phone:

For assistance or questions regarding the Energy Standards, contact the Energy Hotline at: 1-800-772-3300

Registration Number:

Registration Date/Time:

HERS Provider:

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CF1R-ALT-01-E User Instructions

Minimum requirements for prescriptive alteration compliance can be found in Building Energy Efficiency Standards Section 150.2(b)1.

Completing these forms will require that you have the Reference Appendices for the ~~2022~~ 2025 Building Energy Efficiency Standards. This document contains the Joint Appendices which are used to determine climate zone and to complete the section for opaque surfaces. When the term CF1R is used it means the CF1R-ALT-01.

Instructions for sections with column numbers and row numbers are given separately.

If any part of the alteration does not comply, prescriptive compliance fails, in which case the performance compliance approach must be used in an attempt to achieve compliance.

A. General Information

1. Project Name: Identifying information, such as owner's name.
2. Date: Date of document preparation.
3. Project Location: Legal street address of property or other applicable identifying information.
4. Building Front Orientation: Building front expressed in degrees, where North = 0, East = 90, South = 180, and West = 270. Indicate cardinal if it is a subdivision built in multiple orientations. The standards (section 100.1) include the following additional details for determining orientation:
 - Cardinal covers all orientations (for buildings that will be built in multiple orientations);
 - North is oriented to within 45 degrees of true north, including 45 degrees east of north;
 - East is oriented to within 45 degrees of true east, including 45 degrees south of east;
 - South is oriented to within 45 degrees of true south, including 45 degrees west of south;
 - West is oriented to within 45 degrees of true west, including 45 degrees south of west.
5. CA City: Legal city/town of property.
6. Number of Altered Dwelling Units: this field will automatically default to 1 for single family.
7. Zip Code: 5-digit zip code for the project location (used to determine climate zone).
8. Fuel Type: ~~Natural Gas, Liquefied Propane Gas, or~~ this field will automatically default to Electricity.
9. Climate Zone: From Reference Appendices, Joint Appendix, JA2.1.1.
10. Total Conditioned Floor Area: Enter the new conditioned floor area in square feet (ft²), as measured from the outside of exterior walls of the dwelling unit or building being altered.
11. Building Type: this field will automatically default to Single Family (includes duplex).
12. Slab Area: Area of the first floor slab (if any) in square feet (ft²).

13. Project Scope: Check all that apply – insulation, roof replacement > 50%, kitchen remodel, space heating system, space cooling system, duct system, water heating, adding fenestration/glazing, replacing fenestration/glazing, adding fenestration/glazing less than or equal to 75 ft² windows, replacing fenestration/glazing less than or equal to 75 ft² window, adding fenestration/glazing less than or equal to 16 ft² skylight and or replacing fenestration/glazing skylights

~~(2) building with at least R-38 ceiling insulation, (3) buildings with a radiant barrier in the attic, where the radiant barrier is not installed directly above spaced sheathing, meeting the requirements of Section 150.1(c)2, (4) buildings that have no ducts in attic in Climate Zones 2, 4, 9, 10, 12, and 14, or (5) R-2 or greater insulation above or below the roof deck.~~

B. Opaque Surface ~~Building Insulation~~ Details - Framed (Section 150.2(b)1)

1. Tag/ID: A label (if any) from the plans, such as A1.4 or wall.
2. Assembly Type: Ceiling, Wall, Floor.
3. Frame Type: Wood or Metal.
4. Frame Depth: Nominal dimensions (in inches) of framing material such as 2x4 or 2x6.
5. Frame Spacing: 16, ~~or 24~~ or 48 inches on center.

6a. Proposed Cavity R-value: Insulation installed between framing.

NOTE: Section 110.8(d) specifies that if adding insulation to an existing attic, the resulting attic insulation must total R-30. However, the amount of insulation required is limited to the amount of room available for insulation without conflicting with Building Code Section 1203.2.

6b. Proposed Continuous Insulation R-value: Insulation installed on the exterior. See Reference Appendices, Joint Appendix, JA4 for guidance.

7. Proposed U-factor: The U-factor for the entire wall, roof, or floor assembly.

~~8. Joint Appendix JA4 Table: Table number used to determine the R-value or U-factor (e.g., an attic assembly is 4.2.1).~~

~~9. Joint Appendix JA4 Cell: Cell number used to determine the R-value or U-factor (e.g., an R-38 ceiling with 24-inch on center framing is A21).~~

~~10.~~ 8. Required U-factor: From mandatory requirements in Sections 110.0 and 150.0.

~~11.~~ 9. Comments: Any notes regarding location or unique conditions.

C. Opaque Surface ~~Building Insulation~~ Details – Non-framed

1. Tag/ID: A label (if any) from the plans, such as A1.4 or wall.
2. Assembly Type: Roof or Wall.
3. Assembly Material: SIP OSB, SIP I-Joist, SIP Single 2x, SIP Double 2x, see JA4 for guidance.
4. Thickness: Thickness in inches.
5. Proposed Core Insulation R-value: Insulation installed within the materials or on the inside. See Reference Appendices, Joint Appendix, JA4 for guidance.
6. Proposed Continuous Insulation R-value: Insulation installed on the exterior. See Reference Appendices, Joint Appendix, JA4 for guidance.

7. Proposed U-factor: Assembly U-factor from Reference Appendices, Joint Appendix, JA4 or CF1R-ENV-02. Must be less than or equal to Column 10.

~~8. Appendix JA4 Table: Table number used to determine the R-value or U-factor (e.g., an Insulating Concrete Form (ICF) wall is 4.3.13).~~

~~9. Appendix JA4 Cell: Cell number used to determine the R-value or U-factor (e.g., an 8-inch thick Insulating Concrete Form (ICF) wall with 2 inches of Expanded Polystyrene (EPS) (R-15.4) is A6).~~

~~10.~~8. Required U-factor from Table 150.1-A: Based on assembly type and climate zone.

~~11.~~9. Comments: Any notes regarding location, unique conditions, or attachments.

D. Opaque Surface Building Insulation Details – Masonry/Mass Walls

1. Tag/ID: A label (if any) from the plans, for example, A1.4 or wall.

2. Walls Above Grade: Yes or No.

3. Masonry/Mass Type: Clay Brick, Clay Hollow Unit, Concrete Masonry Unit (CMU) Light Weight, Concrete Masonry Unit (CMU) Medium Weight, Concrete Masonry Unit (CMU) Normal Weight, Concrete, Insulating Concrete Form (ICF). See Reference Appendices, Joint Appendix, JA4 for guidance.

4. Masonry/Mass Thickness: Thickness (in inches) of mass.

~~5. Mass Wall Joint Appendix JA4 Reference Table: Table number used to determine the R-value or U-factor (e.g., an Insulating Concrete Form (ICF) wall is 4.3.13).~~

~~6. Mass Wall Joint Appendix JA4 Reference Cell: Cell number used to determine the R-value or U-factor (e.g., an 8-inch thick Insulating Concrete Form (ICF) wall with 2 inches of Expanded Polystyrene (EPS) (R-15.4) is C1).~~

~~75-86.~~ Proposed Exterior Insulation R-value or U-factor: Enter the R-value or U-factor of proposed insulation on the outside surface of the mass wall. See Reference Appendices, Joint Appendix, JA4 for guidance. Use the same descriptor (R-value or U-factor) throughout Table D.

~~97-108.~~ Proposed Interior Insulation R-value or U-factor: Enter the R-value or U-factor of proposed insulation on the inside surface of the mass wall. See Reference Appendices, Joint Appendix, JA4 for guidance. Use the same descriptor (R-value or U-factor) throughout Table D.

~~11. Insulation Layer Joint Appendix JA4 Table: Table number used to determine the R-value or U-factor (e.g., an Insulating Concrete Form (ICF) wall is 4.3.13).~~

~~12. Insulation Layer Joint Appendix JA4 Cell: Cell number used to determine the R-value or U-factor (e.g., an 8-inch thick Insulating Concrete Form (ICF) wall with 2 inches of Expanded Polystyrene (EPS) (R-15.4) is A6).~~

~~139-1410.~~ Required Exterior Insulation R-value or U-factor: The required R-value or U-factor (whichever descriptor was selected in Column ~~75~~ or ~~86~~) for exterior insulation will be completed based on the Table 150.1-A requirements for the wall type.

~~1511-1612.~~ Required Interior Insulation R-value or U-factor: The required R-value or U-factor (whichever descriptor was selected in Column ~~97~~ or ~~10~~) for interior insulation will be completed based on the Table 150.1-A requirements for the wall type.

E. Roof Replacement (Section 150.2(b)~~1H1I~~)

When 50% or more of the roof is being replaced the roofing requirements are triggered. Any areas of roof covered by building integrated photovoltaic panels and solar thermal panels are exempt; however, the area of roof not covered by photovoltaic panels would still need to meet any applicable cool roof requirements. Additionally, there are many alternatives/exceptions when a cool roof is required.

When the roof is steep-sloped (pitch greater than or equal to 2:12) the roof requirements include a cool roof in climate zones 4 and 8-15. The minimum requirement is 0.20 Aged Solar Reflectance, 0.75 Thermal Emittance, or a minimum SRI of 16.

1. Tag/ID: A label, if any, from the plans, for example R-1.
2. Method of Compliance: Indicate if the method of compliance is going to be based on Aged Solar Reflectance and Thermal Emittance, the Solar Reflectance Index (SRI), or an Exception.
3. Roof Pitch: Expressed as 4:12, for example, which means the roof rises 4 feet within a span of 12 feet. When roofs have multiple pitches the requirements are based on the pitch of 50% or more of the roof.
4. Exception: If meeting one of the exceptions. Indicate which exception is, or will be, met.

NOTE: Exceptions and alternatives for steep slope roofs:

- (a) Mass roof 25 pounds per square foot (lbs/ft²) or greater (such as sod roof);
- (b) Roof has ceiling assemblies with a U-factor less than or equal to 0.025 or R-38 insulation;
- (c) Roof has a radiant barrier not installed directly above spaced sheathing meeting 150.1(c)2;
- (d) R-2 continuous insulation above or below the roof deck.

In climate zones 4 and 6-15, when there is a low-sloped roof (pitch less than 2:12) the cool roof requirements are for a minimum Aged Solar Reflectance of 0.63, a minimum 0.75 Thermal Emittance, or a minimum SRI of 75.

NOTE: Exceptions and alternatives for low slope roofs:

- (a) Mass roof 25 pounds per square foot (lbs/ft²) or greater (uncommon situation such as sod roof);
- (b) Roof deck installation trade off—by installing roof deck insulation, a lower aged solar reflectance is required: In Climate Zones 6 and 7 R-2 (0.60), R-4 (0.55), R-6 (0.50), R-8 (0.45), R-10 (no requirement); In Climate Zones 2, 4 and 8-15 R-16 (0.60), R-18 (0.55), R-20 (0.50), R-24 (no requirement).

NOTE: If one of the exceptions above has been selected then the rest of Section E is not required.

5. The CRRC Product ID Number is obtained from the [Cool Roof Rating Council's Rated Product Directory](#). Products are listed by manufacturer, brand, type of installation, roofing material, and color, as well as product performance.
6. Product Type: See [Cool Roof Rating Council's directory](#). Generally product types include single-ply roof, wood shingles, asphalt roof, metal roof, tile roof.
7. R-value Deck Insulation: If one of the exceptions selected includes adding roof deck insulation, indicate the R-value of insulation.
8. Proposed Initial Solar Reflectance: Based on the product chosen from the [Cool Roof Rating Council's Rated Product Directory](#). If using default assumption indicate NA since the Aged Solar Reflectance is available.
9. Proposed Aged Solar Reflectance: Value is from the [Cool Roof Rating Council's Rated Product Directory](#). If the aged value is not available, calculate the calculated Aged Solar Reflectance using the Solar Reflectance Index (SRI) Calculation worksheet located on the [California](#)

[Energy Commission website](#) or the aging equation $p_{aged} = [0.2 + \beta(p_{initial} - 0.2)]$, where $p_{initial}$ = the initial solar reflectance and soiling resistance β is listed by product type below.

VALUES OF SOILING RESISTANCE β BY PRODUCT TYPE

Product Type	CRRC Product Category	β
Field-Applied Coating	Field-Applied Coating	0.65
Other	Not A Field-Applied Coating	0.70

10. Proposed Thermal Emittance: From the product specification default value. If using a calculated Solar Reflectance Index (SRI) place the Thermal Emittance used to calculate SRI.
11. Proposed Solar Reflectance Index (SRI): It is optional to meet the SRI but if chosen to do so, use the Solar Reflectance Index (SRI) Calculation Worksheet found on the [California Energy Commission website](#).
12. Minimum Required Aged Solar Reflectance: Based on climate zone and roof slope.
13. Minimum Required Thermal Emittance: Based on climate zone and roof slope.
14. Minimum Required Solar Reflectance Index (SRI): Based on climate zone and roof slope.

NOTE: If the cool roofing requirements will be met by a liquid field applied coating, Section 110.8(i)4 requires the coating be applied across the entire roof surface and meet the dry mil thickness or coverage recommended by the manufacturer.

F. Fenestration/Glazing Allowed Areas and Efficiencies (Section 150.2(b)1)

The climate zone and scope of the alteration will affect the amount of fenestration (also known as glazing) allowed. If limited to 20 percent, this is calculated as Conditioned Floor Area x 0.20 = total square footage (ft²) of fenestration allowed (20 percent). Fenestration areas are expressed in feet, not inches. When west-facing fenestration is limited (in climate zones 2, 4, and 6-~~16~~¹⁵), it is limited to a maximum of 5 percent. Additions of 1,000 square feet (ft²) or less have alternate requirements. For example, the limit may be 120 square feet (ft²) of fenestration or 25 percent. While west-facing fenestration may be limited, if there is no west fenestration the upper limit remains at 120 square feet (ft²) or 25 percent (or the values shown in columns 2 and 3).

1. Alteration Type: Auto-filled with the project scope in A13: adding fenestration/glazing, replacing fenestration/glazing, ~~adding fenestration/glazing less than or equal to 75 square feet (ft²) windows,~~ replacing fenestration/glazing less than or equal to 75 square feet (ft²) window, adding fenestration/glazing less than or equal to 16 square feet (ft²) skylight and or replacing fenestration/glazing skylights.

2. Maximum Allowed Fenestration Area for All Orientations (ft²): The maximum total fenestration area is 20 percent. Depending on the type of fenestration and the alteration type, this field may show values such as 75 square feet (ft²).
3. Maximum Allowed West-Facing Fenestration Area Only: Calculated value based on Conditioned Floor Area multiplied by 5 percent (Used in climate zones 2, 4, and 6-16)

NOTE: (1) If adding fenestration/glazing less than or equal to 16 square feet (ft²) skylight, enter NA

(2) West includes any vertical fenestration oriented to within 45 degrees of true west, including 45 degrees south of west. For skylights, west also includes any skylight area facing any direction with a pitch of less than 1:12

4. Existing Fenestration Area for All Orientations: Enter the area, in square feet (ft²), of the existing fenestration/glazing.
Existing West-Facing Fenestration Area: Enter the area, in square feet (ft²), of the existing west-facing fenestration/glazing. If project has no existing west-facing fenestration then enter "0".

5. Maximum Allowed U-factor: Maximum U-factor from Package A or Table 150.1-A. ~~This field will almost always be 0.27 or 0.30, depending on climate zone. For skylights this will be 0.55/0.40.~~

~~NOTE: (1) If meeting Exception 2 to Section 150.2(b)1A (adding less than or equal to 16 square feet (ft²) of new fenestration or skylights), shall not be required to meet the total fenestration area and west facing fenestration area requirements of Sections 150.1(c)3B and C, enter 0.55.~~

~~(2) If meeting Exception 1 to Section 150.2(b)1B (replacing less than or equal to 75 square feet (ft²) windows), enter 0.40.~~

~~(3) If meeting Exception 2 to Section 150.2(b)1B (replacing skylights), enter 0.40.~~

6. Maximum Allowed SHGC: Maximum SHGC from Package A or Table 150.1-A. ~~This field will almost always be either 0.23 or N/A, depending on climate zone or 0.20 in climate zone 15. N/A means there is no maximum SHGC required in this climate zone. For skylights this will almost always be either 0.27 or 0.30.~~

NOTE:

~~(1) If meeting Exception 1 to Section 150.2(b)1B, (replacing less than or equal to 75 square feet (ft²) windows), enter 0.35 in Climate Zones 2, 4, and 6- through 15.~~

~~(2) If meeting Exception 2 to Section 150.2(b)1B (replacing skylights), enter 0.30.~~

~~(3) If meeting Exception 3 to Section 150.2(b)1B (vertical fenestration shall have a maximum SHGC value no greater than 0.23 in Climate Zone 15-), enter no greater than 0.23.~~

7. Comments: Note any special location or comment here.

G. Fenestration/Glazing Proposed Areas and Efficiencies – Add (Section 150.2(b)1A)

1. Tag/ID: A label (if any) from the plans, such as W1.
2. Fenestration Type: Indicate the type of fenestration construction (e.g., Fixed Window, Operable Window, or Skylight).
NOTE: Doors with glazing are counted in one of two ways. A door with 25 percent or more glazing is counted as the entire door area. A door with less than 25 percent glazing can be counted as the entire door area or can be calculated as the actual glass area with a 2-inch frame all around.
3. Frame type: Metal, metal thermal break, or non-metal.
4. Dynamic Glazing: Indicate if the fenestration has integrated shading device, chromogenic glazing, or none for no dynamic glazing. Chromogenic glazing shall be considered separately from other fenestration types.
5. Orientation (North, East, South, West). In climate zones where the West-facing glazing is limited, list west-facing individually. The definitions in the Energy Standards include these specific details:
 - North is oriented to within 45 degrees of true north, including 45 degrees east of north;
 - East is oriented to within 45 degrees of true east, including 45 degrees south of east;
 - South is oriented to within 45 degrees of true south, including 45 degrees west of south;
 - West is oriented to within 45 degrees of true west, including 45 degrees north of west.
 NOTE: Skylights in a roof pitch greater than 1:12 can be included as facing the same orientation as that portion of the roof angle. If the skylight is in a roof with a pitch less than 1:12, the skylight is assumed to face west.
6. Number of Panes: Indicate the number of panes for each Tag/ID; is it single, double, or triple pane window?
7. Proposed Fenestration Area (ft²): Indicate the area in square feet (ft²) of each exterior fenestration type, excluding west-facing fenestration.
8. Proposed West Facing Fenestration Area (ft²): In climate zones 2, 4, 6-~~16~~15, indicate the area (in ft²) of each exterior west-facing fenestration type separately.
NOTE: Skylights installed in a roof with pitch less than 1:12 are considered to face west.
9. Proposed U-factor: Enter
 - (a) the National Fenestration Rating Council (NFRC) U-factor based on the proposed brand and type of fenestration using [NFRC certified values](#); or
 - (b) the default value from Table 110.6-A; or
 - (c) the NA6.2 alternate default U-factor (for non-rated site-built fenestration only); or
 - (d) the Area-weighted Average from CF1R-ENV-02.

If any products (other than skylights) have a higher U-factor than ~~0.27 or 0.30, depending on climate zone~~required by Package A, first complete a CF1R-ENV-02 to calculate the Area-Weighted Average U-factor, and attach it to the CF1R-ALT-01.

NOTE: Dynamic glazing is a glazing system that changes its performance U-factor and SHGC based on the physical environment. Dynamic glazing includes chromogenic glazing or integrated shading systems (this does not include internally or externally mounted shading devices). If using dynamic glazing, use the lowest tested U-factor and SHGC in Columns 9 and 11.

10. Source: National Fenestration Rating Council (NFRC), Table 100.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-Weighted Average Worksheet (CF1R-ENV-02). The source of the U-factor data for the fenestration product.

11. Proposed SHGC: In climate zones 2, 4, 6-~~16~~ 15 enter

- (a) the National Fenestration Rating Council (NFRC)- solar heat gain coefficient (SHGC) based on the proposed brand and type of fenestration using [NFRC certified values](#); or
- (b) the default value Table 110.6-B; or
- (c) the NA6.3 alternate default solar heat gain coefficient (SHGC) (for non-rated site-built fenestration only); or
- (d) the Area-weighted Average from CF1R-ENV-02.

If any products (other than skylights) have a higher solar heat gain coefficient (SHGC) than required by Package A, first complete a form CF1R-ENV-02 to calculate the Area-Weighted Average SHGC and attach it to the CF1R-ALT-01.

12. Source: National Fenestration Rating Council (NFRC), Table 100.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-Weighted Average Worksheet (CF1R-ENV-02). The source of the solar heat gain coefficient (SHGC) data for the fenestration product.

13. Exterior Shading Device: If exterior shading devices are used to meet the solar heat gain coefficient (SHGC) requirement, indicate the type of device (from Table S-1 of CF1R-ENV-03 Solar Heat Gain Coefficient Worksheet) and attach an ENV-03.

NOTES:(1) An exterior shading device is not used for products with a National Fenestration Rating Council (NFRC) rated U-factor and solar heat gain coefficient (SHGC) based on a factory integrated shading device.

(2) Chromogenic glazing shall be considered separately from other fenestration.

(3) If using an overhang for south-facing glazing, the glazing must be fully shaded at solar noon on August 21 and substantially exposed to direct sunlight at solar noon on December 21 (see Residential Compliance Manual, Section 3.5.5).

14. Combined SHGC from CF1R-ENV-03: If exterior shading devices are combined with the solar heat gain coefficient (SHGC) value of the fenestration to meet the prescriptive SHGC requirements (as indicated by a value in Column E. 13), indicate the SHGC calculated on compliance document CF1R-ENV-03 and attach the one for each window with an exterior shading device.

15.-32. Automatically completed entries; no user input required.

H. Fenestration/Glazing Proposed Areas and Efficiencies – Replace (Section 150.2(b)1B)

- 1. Tag/ID: A label (if any) from the plans, such as W1.
- 2. Fenestration Type: Indicate the type of fenestration construction (e.g., Fixed Window, Operable Window, or Skylight).

NOTE: Doors with glazing are counted in one of two ways. A door with 25 percent or more glazing is counted as the entire door area. A door with less than 25 percent glazing can be counted as the entire door area or can be calculated as the actual glass area with a 2-inch frame all around.

- 3. Frame type: Metal, metal thermal break, or non-metal.

4. Dynamic Glazing: Indicate if the fenestration has integrated shading device, chromogenic glazing, or none for no dynamic Glazing. Chromogenic glazing shall be considered separately from other fenestration types.
5. Orientation (North, East, South, West). In climate zones where the West-facing glazing is limited, list west-facing individually. The definitions in the Energy Standards include these specific details:
 - North is oriented to within 45 degrees of true north, including 45 degrees east of north;
 - East is oriented to within 45 degrees of true east, including 45 degrees south of east;
 - South is oriented to within 45 degrees of true south, including 45 degrees west of south;
 - West is oriented to within 45 degrees of true west, including 45 degrees north of west.

NOTE: Skylights in a roof pitch greater than 1:12 can be included as facing the same orientation as that portion of the roof angle. If the skylight is in a roof with a pitch less than 1:12, the skylight is assumed to face west.

6. Area Removed (ft²): Enter the area, in square feet (ft²), of the fenestration/glazing being removed.
7. Area Added (ft²): Enter the area, in square feet (ft²), of the fenestration/glazing being added.
8. Net Added Area (ft²): The difference between the Area Added and the Area Removed.
9. Proposed U-factor: Enter
 - (a) the National Fenestration Rating Council (NFRC) U-factor based on the proposed brand and type of fenestration using [NFRC certified values](#); or
 - (b) the default value from Table 110.6-A; or
 - (c) the NA6.2 alternate default U-factor (for non-rated site-built fenestration only); or
 - (d) the Area-Weighted Average from CF1R-ENV-02.

If any products (other than skylights) have a higher U-factor than ~~0.30~~ [required by Package A](#), first complete a CF1R-ENV-02 to calculate the Area-Weighted Average U-factor and attach it to the CF1R-ALT-01.

NOTE: Dynamic glazing is a glazing system that changes its performance U-factor and SHGC based on the physical environment. Dynamic glazing includes chromogenic glazing or integrated shading systems (this does not include internally or externally mounted shading devices). If using dynamic glazing, use the lowest tested U-factor and SHGC in Columns 9 and 11.

10. Source: National Fenestration Rating Council (NFRC), Table 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-weighted Average Worksheet (ENV-02). The source of the U-factor data for the fenestration product.
11. Proposed Solar Heat Gain Coefficient (SHGC): In climate zones 2, 4, 6-~~16~~ [through 15](#) enter
 - (e) the National Fenestration Rating Council (NFRC)- solar heat gain coefficient (SHGC) based on the proposed brand and type of fenestration using [NFRC certified values](#), or

- (f) the default value Table 110.6-B, or
- (g) the NA6.3 alternate default solar heat gain coefficient (SHGC) (for non-rated site-built fenestration only), or
- (h) the Area-weighted Average from CF1R-ENV-02.

If any products (other than skylights) have a higher SHGC than required by Package A, first complete a form CF1R-ENV-02 to calculate the area-weighted average SHGC and attach it to the CF1R-ALT-01.

- 12. Source: National Fenestration Rating Council (NFRC), Table 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-weighted Average Worksheet (ENV-02). The source of the SHGC data for the fenestration product.
- 13. Exterior Shading Device: If exterior shading devices are used to meet the SHGC requirement, indicate the type of device (from Table S-1 of CF1R-ENV-03 Solar Heat Gain Coefficient Worksheet) and attach an ENV-03.

NOTES: (1) An exterior shading device is not used for products with a National Fenestration Rating Council (NFRC) rated U-factor and SHGC based on a factory integrated shading device.

(2) Chromogenic glazing shall be considered separately from other fenestration.

(3) If using an overhang for south-facing glazing, the glazing must be fully shaded at solar noon on August 21 and substantially exposed to direct sunlight at solar noon on December 21 (see Residential Compliance Manual, Section 3.5.5).

- 14. Combined Solar Heat Gain Coefficient (SHGC) from CF1R-ENV-03: If exterior shading devices are combined with the SHGC value of the fenestration to meet the prescriptive SHGC requirements (as indicated by a value in column F. 13), indicate the SHGC calculated on compliance document CF1R-ENV-03 and attach the form for each window with an exterior shading device.
- 15.-30. Automatically completed entries; no user input required.

H1.1. Opaque Swinging Doors to Exterior

- 1. Tag/ID: Provide a name or designator for each unique door. This designator should be used consistently throughout the plan set (elevations, door schedules, etc.)
- 2. Area: Calculated area in square feet (ft²) for each unique door.
- 3. Proposed U-factor: Enter the proposed U-factor. If value is greater than 0.20, column 06 will autocomplete as Yes.
- 4. Source: National Fenestration Rating Council (NFRC) or Reference Appendices, Joint Appendix, Table 4.5.1s, 110.6-A and 110.6-B, Equations NA6-1 and NA6-2, or Area-Weighted Average Worksheet (CF1R-ENV-02).
- 5. Required Maximum U-factor. This field will always be 0.20.
- 6. Weighted Average: If column 03 is greater than 0.20 U-factor, attach form CF1R-ENV-02-E.
- 7. Comments: Any notes regarding location, unique conditions, or attachments.

J. Space Conditioning (SC) Systems – Heating/Cooling (Section 150.2(b))

Requirements of the standards apply to a heating and cooling system alteration based on the type of alteration and the system type (Section 150.2(b)1). A completely new system will meet all mandatory and prescriptive requirements, which vary by climate zone (based on Section 150.2(b)1C).

NOTE: Computer performance compliance can be used to trade-off any requirements that are not mandatory. When parts of a system are replaced, it may trigger some of the same requirements that apply to new systems and duct alterations. A Certificate of Compliance for Alterations to Space Conditioning Systems (CF1R-ALT-02) is required for each dwelling unit with a space conditioning system alteration.

1. Dwelling Unit Name: Name of dwelling unit or any other identifying name.
2. Dwelling Unit Total CFA (ft²): Conditioned Floor Area in square feet (ft²), as measured from the outside of exterior walls of the dwelling unit or building being altered.
3. Comments: Any notes regarding location or unique conditions.

K. Water Heating Systems (Section 150.2(b)1H)

Water heating compliance for an alteration is described in Section 150.2(b)1H.

1. Is natural gas connected to the existing water heater? ~~Is the existing water heater electric resistance?~~ Yes or No
2. Water Heating System Identification or Name: Provide a unique name for each unique water heating system type in the building. If the same water heating system type is used in more than one location in the building, it is sufficient to list the unique water heating system type only once.
- ~~3. Water Heating System Type: Domestic Hot Water (DHW), Hydronic, or Combined Hydronic. DHW is for domestic hot water, hydronic is a water heating system used for space heating only; combined hydronic are when the water heater will provide both space conditioning and domestic hot water.~~
- ~~4.~~3. System option:
 1. A natural gas or propane water heating system.
 2. A single heat pump water heater, storage tank shall not be located outdoors and placed on a incompressible, rigid insulated surface with a minimum thermal resistance of R-10
 3. A single Tier 3 or higher heater (as rated by Northwest Energy Efficiency Alliance (NEEA).
 4. If no natural gas is connected to the existing water heater location, a consumer electric water heater (Heat Pump Water Heater or Electric Resistance)
- ~~4.~~5. If the existing water heater is electric resistance, a consumer electric water heater
- ~~5.~~4. Water heater Type: Consumer instantaneous, consumer storage, NEEA Tier 3 or higher heat pump water heater
- ~~6.~~ Volume (gal): Tank capacity in gallons. For instantaneous water heaters, enter N/A.
- ~~7.~~5. Fuel Type: Gas, Propane. heat pump, electricity.

~~8~~6. _____ Number of water heaters in system: No more than 1 per dwelling unit allowed.

Documentation Declaration Statements

1. The person who prepared the CF1R will sign and complete the fields for their name, company (if applicable), address, phone number, certification information (if applicable), date and signature.
2. The person who is assuming responsibility for the project being built to comply with Title 24, Part 6, will complete the fields for their name, company (if applicable), address, phone number, license number (if applicable), date and signature.

Registration

1. The CF1R must be registered with a ~~HERS~~ECC provider prior to submitting for a building permit. See Residential Manual Section 2.1.1.

For information and data collection only. Not valid until registered with an ECC provider

A. General Information

01	Project Name:	<<User Input: Text>>	02	Date Prepared:	<<User Input: Date>>
03	Project Location:	<<User Input: String>>	04	Building Front Orientation (deg or cardinal):	<<User Input: IntegerNonnegative>>
05	CA City:	<<User Input: String>>	06	Number of Altered Dwelling Units:	<<User Input: IntegerNonnegative, Fixed value = 1>>
07	Zip Code:	<<User Input: Zipcode>>	08	Fuel Type:	<<User selects from list: Fixed value = Electricity, Natural Gas, Propane>>
09	Climate Zone:	<<User selects from list: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16>>	10	Total Conditioned Floor Area (ft²):	<<User Input: IntegerNonnegative>>
11	Building Type:	<< Fixed value = user-select one from list: *Single Family >>	12	Slab Area (ft²)	<<User Input: IntegerNonnegative>>
13	Project Scope:	<< user select as many as are applicable from list: *Insulation (Show tables B, C & D) *Roof Replacement >50% (Show table E and B) *Kitchen Range Hood installation (new or replacement) *Space Heating System (display Table I) *Space Cooling System (display Table I) *Space Conditioning Duct System (display Table I) *Water Heating (Table J) <<User may select 2 from the following: *Adding Fenestration/Glazing (Show tables F and G, and if selected do not allow user to also select Adding Fenestration/Glazing ≤ 75 ft² Windows) *Adding Fenestration/Glazing ≤ 75 ft² Windows (Show tables F and G, and if selected do not allow user to also select Adding Fenestration/Glazing) *Adding Fenestration/Glazing ≤16 ft² Skylight (Show tables F and G)>> <<User may select 2 from the following: *Replacing Fenestration/Glazing (Show tables F and H, and if selected do not allow user to also select Replacing Fenestration/Glazing ≤ 75 ft²) *Replacing Fenestration/Glazing ≤ 75 ft² Windows (Show tables F and H, and if selected do not allow user to also select Replacing Fenestration/Glazing) *Replacing Fenestration/Glazing Skylight (Show tables F and H)>> *Opaque Exterior Doors>>			
Note: <<Static Text - see the top>>					

B. Opaque Surface Building Insulation Details - Framed (Section 150.2(b)1)

<<if A13 = Insulation then display this section; else display standard “This Section Does Not Apply” message>>

01	02	03	04	05	06a	06b	07	08	09	1008	1109
Tag/ID	Assembly Type	Frame Type	Frame Depth (inches)	Frame Spacing (inches)	Proposed					Required	Comments
					Cavity R-value	Continuous Insulation R-value	U-factor	Joint Appendix JA4 Reference		U-Factor or R-value	
								Table	Cell		
<<User Input: ObjectNamePermissible>>	<<User selects from list: Ceiling, Floor, Roof, Wall or Attic>>	<<User selects from Wood or Metal>>	<<User selects from 2x4, 2x6, 2x8, 2x10, 2x12 or 2x14>>	>> user selects from list: 16 inches on center, 24 inches on center, 48 inches on center>>	<<User Input: DecimalNonnegative>>	<<User Input: DecimalNonnegative>>	<<User Input: DecimalNonnegative>>	<<User Input: JA4 Table ID>>	<<User Input: JA4 Table Cell>>	<<User Input: DecimalNonnegative>>	<<User Input: Text>>

Note:

<<Static Text - see the top>>

C. Opaque Surface Building Insulation Details – Nonframed

<<if A13 = Insulation then display this section; else display standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	1008	1109
Tag/ID	Assembly Type	Assembly Materials	Thickness (inches)	Proposed					Required	
				Core Insulation R-value	Continuous Insulation R-value	U-factor	Joint Appendix JA4 Reference		Required	Comments
							Table	Cell		
<<user input: ObjectNamePermissive>>	<<user selects from list: Roof, Wall>>	<<user selects from list: *SIP OSB, *SIP 2x, *SIP 4x, *SIP I-joist, *Metal Panel Walls, *Log Home Walls, *Straw Bale Walls, *Insulating Concrete Form >>	<<user input: DecimalNonnegative>>	<<user input: DecimalNonnegative>>	<<user input: DecimalNonnegative>>	<<user input: DecimalNonnegative>>	<<use input: JA_TableID>>	<<use input: JA_TableCell>>	<<user input: DecimalNonnegative>>	<<user input: Text>>

D. Opaque Surface Building Insulation Details – Masonry/Mass Walls

<<if A13 = Insulation then display this section; else display standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	0705	0806	0907	1008	11	12	1309	1410	1511	1612
Tag/ID	Above or Below Grade?	Masonry/ Mass Wall Type	Masonry/ Mass Wall Thickness (inches)	Proposed								Required			
				Mass-Wall Joint Appendix JA4 Reference		Exterior Insulation		Interior Insulation		Insulation Layers Joint Appendix JA4 Reference		Exterior Insulation		Interior Insulation	
				Table	Cell	R-value	U-factor	R-value	U-factor	Table	Cell	R-value	U-factor	R-value	U-factor
<<User input: ObjectNamePermissive>>	<<User select from list: *Above Grade; or *Below Grade>>	<<User selects from list: *Clay Brick, *Clay Hollow Unit, *CMU Light Weight, *CMU Medium Weight, *CMU	<<User input: DecimalNonnegative>>	<<User input: JA_TableID>>	<<User input: JA_TableCell>>	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same	<<User input: JA_TableID>>	<<User input: JA_TableCell>>	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same	<<User input: DecimalNonnegative; Note: Either R-value or U-factor is required, not both. The same

		Normal Weight, *Concrete				descript or should be used throughout this table>>	r should be used throughout this table>>	r should be used throughout this table>>	t this table>>			descript or should be used throughout this table>>	t this table>>	r should be used throughout this table>>	

Note:
<<Static Text - see the top>>

E. Roof Replacement (Section 150.2(b)~~1H~~1I)

<<if A13 is Roof Replacement> 50% then display this section; else display section header and standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07	08	09	10	11	12	13	14
Tag/ ID	Method of Compliance	Roof Pitch	Exception	CRRC Product ID Number	Product Type	R-value Deck Insulation	Proposed				Minimum Required		
							Initial Solar Reflectance	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)	Aged Solar Reflectance	Thermal Emittance	SRI (Optional)
<<User Input: Object Name Per missiv e>>	<<User selects from dropdown, Not in an applicable climate zone, Aged Solar Reflectance and Thermal Emittance, Initial Solar Reflectance and Thermal Emittance, SRI or Exception>>	<< User selects from drop- down: less than 2:12 or greater than or equal to 2:12 >>	<< User selects from dropdown, If cell 02 equals Exception and cell 03 is greater than or equal to 2:12 and A09 = 4, or 8-15, then choices are: Mass roof ≥ 25 lbs/ft2, Roof area covered by building integrated PV panels or solar thermal panels,	<<if E03 = greater than or equal to 2:12 and E02 = exception , then report NA; Else if E03 = less than 2:12, E02 = exception and E04 ≠ roof deck installatio n, then report NA; Else user input	<<if cell 02 equals exception, cell 03 equals less than 2:12 and cell 04 does not equal roof deck installatio n return NA; Else user selects from dropdown : Asphalt Shingles, Built-up Roofing, Concrete/ Clay Roof	<<if cell 03 equals less than 2:12 and cell 04 equals insulation above the roof deck user selects from dropdown: R-2, R-4, R-6, R- 8, R-12, R- 16, R-20, R-24; Else N/A>>	<<if cell 02 equals Initial Solar Reflectance and Thermal Emittance User input value from the CRRC Directory; Elseif cell 02 equals Exception, cell 03 equals less than 2:12 and cell 04 equals roof deck installation return NA; Elseif cell 02 equals Initial Solar Reflectance and Thermal Emittance user input	<<if cell 02 = ‘Not in an applicable climate zone’ or ‘SRI’ return NA; Else if cell 02 equals exception, cell 03 equals less than 2:12 and cell 04 does not equal roof deck installation return NA; Elseif cell 02 equals Initial Solar Reflectance and Thermal Emittance user input	<<if cell 02 = ‘Not in an applicable climate zone’ then result = NA; if cell 02 equals exception, cell 03 equals less than 2:12 and cell 04 does not equal roof deck installation return NA, Else user input value from CRRC Directory>>	<< if cell 02 equals SRI, user input value from CRRC Directory or the SRI worksheet Else NA>>	<<if cell 02 equals exception, cell 03 equals less than 2:12, cell 04 equals roof deck insulation, Then-and-If cell 07 equals R-2 return 0.60; Else if cell 02 equals exception, cell 03 equals less than 2:12, cell 04 equals roof deck	<< if cell 02 equals exception and cell 04 does not equal roof deck installation return NA; Elseif cell 02 equals ‘Not in an applicable climate zone’ or ‘SRI’ return NA; Else return 0.75>>	<< if cell 02 equals SRI, cell 03 equals less than 2:12 and cell A09 equals 13 or 15 return 75; Elseif cell 02 equals SRI, cell 03 equals greater than or equal to 2:12 and cell A09 equals 10-15 return 16; Else return NA>>

			Roof has U-factor of ≤ 0.025 or R-38 insulation, Roof has a radiant barrier not installed above spaced sheathing meeting 150.1(c)2, R-2 insulation above the roof deck; if A09 = 2, 4, 9, 10, 12 or 14, then also allow: No ducts are installed in the attic;	value from the CRRC Directory >>	Tiles, Slate, Factory Applied Coating, Field Applied Coating, Metal Roof, Modified Bitumin Sheet, Roof Pavers, Single Ply Thermoplastic, Single Ply Thermoset, Metal Shakes Shingles, Fluid Applied Membrane, Polymer Composite Steep Slope, Spray Polyurethane Foam, Stone Aggregate Ballast>>			calculated value using equation from 110.8(i)2; Else user input value from CRCC Directory>>			insulation and cell 07 equals R-4 return 0.55; Elseif cell 02 equals exception, cell 03 equals less than 2:12, cell 04 equals roof deck insulation and cell 07 equals R-6 return 0.50; Elseif cell 02 equals exception, cell 03 equals less than 2:12, cell 04 equals roof deck insulation and cell 07 equals R-8 return 0.45; Elseif cell 02 equals exception, cell 03 equals less than 2:12, cell 04 equals roof deck insulation and cell 07 equals R-12 return 0.40;		
--	--	--	--	----------------------------------	---	--	--	--	--	--	--	--	--

			thermal panels; Else NA>>								Elseif cell-02 equals exception, cell-03 equals less than 2:12, cell-04 equals roof deck insulation and-cell 07 equals R-16 return 0.35; Elseif cell-02 equals exception, cell-03 equals less than 2:12, cell-04 equals roof deck insulation and-cell 07 equals R-20 return 0.30; Elseif cell-02 equals exception, cell-03 equals less than 2:12, cell-04 equals roof deck insulation and-cell 07 equals R-24 return 0.25; <u>End If</u>		
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											Elseif cell 02 equals Aged Solar Reflectance and Thermal Emittance <u>OR Initial Solar Reflectance and Thermal Emittance</u> And and cell 03 equals less than 2:12 and cell A09 equals 13 or 15 return 0.63; <u>End If</u> Elseif cell 02 equals Initial Solar Reflectance and Thermal Emittance <u>OR Initial Solar Reflectance and Thermal Emittance</u> And and cell 03 equals less than 2:12 and cell A09 equals 13 or 15 return 0.63; <u>End If</u> Elseif cell 02 equals Aged Solar Reflectance and Thermal Emittance		
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F. Fenestration/Glazing Allowed Areas and Efficiencies (Section 150.2(b)1A)

<<if A13= Add Fen./Glazing, Replace Fen./Glazing, ~~Add Fen./Glazing ≤ 75 ft2 Windows~~, Replace Fen./Glazing ≤ 75 ft2 Windows, Add Fen./Glazing ≤ 16 ft2 Skylights, Replace Fen./Glazing Skylight then display this section; else display section header and standard “This Section Does Not Apply” message>>

01	02	03	04a	04b	05a	05b	06a	06b	07
Alteration Type	Maximum Allowed Fenestration Area for All Orientations (ft²) (Windows)	Maximum Allowed West-Facing Fenestration Area Only (ft²)	Existing Fenestration Area for All Orientations (ft²)	Existing West-Facing Fenestration Area (ft²)	Maximum Allowed U-factor (Windows)	Maximum Allowed U-factor (Skylights)	Maximum Allowed SHGC (Windows)	Maximum Allowed SHGC (Skylights)	Comments
<<This value is auto-filled with the Project Scope in section A13: *Add Fen./Glazing, *Replace Fen./Glazing, *Add Fen./Glazing ≤ 75 ft2 Windows , *Replace Fen./Glazing ≤ 75 ft2 Windows, *Replacing Fenestration/ Glazing Skylight, *Adding Fenestration/ Glazing ≤ 16ft2 skylight >>	<<Calculated value: if F01 contains Add Fenestration /Glazing then value equals (A10 x 0.20); else report NA>>	<<Calculated value: if F01 contains Add Fenestration/ Glazing then value equals (A10 x 0.05); Else report NA>>	<<User input: IntegerNonnegative>>	<<User input: IntegerNonnegative>>	<<If F01 contains Replace Fenestration/ Glazing ≤ 75 ft2 Windows then the value equals 0.40; Else if report 0.27 A09 = 1 through 5 and 11 through 14 and 16 then report 0.27; Else if A09 value = 0.30 Or report 0.30 if A09 = 6 through 10, and 15>>	<<if F01 contains Add Fenestration/ Glazing ≤ 16ft2 Skylight, or Replace Fenestration/ Glazing Skylight then value equals 0.40; Else if A09 equals 1 through 5, 11 through 14 and 16 then report 0.27; Else if A09 equals 1 through 5, 11 through 14 and 16 then report 0.27; Else value = 0.30 Else report 0.30>>	<<if A09 = 1, 3, 5 or 16, then result equals NA; If F01 contains Replace Fenestration /Glazing ≤ 75 ft² Windows, then value equals 0.35; If A09 = 15 then value = 0.20; Else value = 0.23 Else report 0.23 if A09 = 2, 4, 6 through 14; Or report 0.20 if A09 = 15>>	<< if A09 = 1, 3, 5 or 16, then result equals NA; if F01 contains Add Fenestration n/Glazing ≤ 16ft2 skylight, or Replace Fenestration n/Glazing Skylight, then value equals 0.30; Else value = 0.23 Else report 0.23 if A09 = 2, 4, 6 through 14; Or report 0.20 if A09 = 15>>	<<User Input: Text>>

G. Fenestration Proposed Areas and Efficiencies – Add (Section 150.2(b)1A)

<<if A13 = Add Fenestration/Glazing, Add Fenestration/Glazing ≤ 75 ft2, or Add Fenestration/Glazing ≤ 16 ft2 then display this section; else display section header and the standard “This section does not apply” message>>

Note: <<Static Text - see the top>>

01	02	03	04	05	06	07	08	09	10	11	12	13	14
Tag/ ID	Fenestration Type	Frame Type	Dynamic Glazing	Orientation N, S, W, E	Number of Panels	Proposed Fenestration Area N, S, E (ft ²)	Proposed West Facing Fenestration Area (ft ²)	Proposed U-factor	Proposed U-factor Source	Proposed SHGC	Proposed SHGC Source	Exterior Shading Device	Combined SHGC from CF1R- ENV-03
<<User Input: ObjectName Permissive> >	<< User selects from list: Fixed Window, Operable Window, and Skylight>>	>> User selects from list: Metal, Metal Thermal Break or non-metal	>> User selects from list: NONE, Chromogeni c Glazing, Integrated Shading device	>> User selects from list: North, East, South and West	>> User selects from list: Single pane, Double pane and Triple pane>>	>> If G05 equals North, East or South, then enter user defined number in this column; else if G05 equals West then value equals NA >>	>> If G05 equals West, then enter user defined number in this column; else if G05 equals North, East, or South then value equals NA>>	<<User Input: DecimalN onnegativ e>>	>> User selects from list: NFRC, Tables 110.6-A and 110.6-B, Equations NA6-1 And NA6-2, or Area- weighted Average Worksheet (ENV- 02)>>	<< If A09 = 1, 3, 5 or 16, then result equals NA Else User Input: DecimalN onnegativ e >>	>> User selects from list: NFRC, Tables 110.6-A and 110.6-B, Equations NA6-1 And NA6-2, or Area- weighted Average Workshee t (ENV-02) Else report NA>>	>> User selects from list: Drop Arm Awnings, Louvered Sun Screen, Low Sun Angle Louvered Sun Screen, None, Operable Awnings, South- facing Overhang, Retractable Awnings, Roll Down Blinds Or Slats, Standard Bug Screen, Sun Screen, Vertical Roller Or Shades>>	>> If A09 = 1, 3, 5 or 16, then result equals NA; Elseif G13 equal to None, or G04 contains Chromog enic glazing, then report N/A; Else report value from CF1R- ENV-03 >>

15	Existing + Proposed Fenestration Area	<<Calculated value: if A13 does NOT contain Add Fenestration/Glazing then report NA; Else [F04a + (sum of G07) + (sum of G08)]>>
16	Maximum Allowed Fenestration Area	<<Report Value from F02>>
17	Compliance Statement	<<If G15 contains NA then value equals NA; If G15 is equal to or less than G16, or G16 equals NA, then report: Design Complies with the Total Allowed Fenestration Area; Else report: Total Proposed Fenestration Area Exceeds Allowable, Do Not Proceed >>
18	Existing + Proposed West-Facing Fenestration Area	<<Calculated value: if A13 does NOT contains Add Fenestration/Glazing then report NA; else [F04a + (sum of G08)]>>
19	Maximum Allowed West Fenestration Area	<<Report Value from F03>>
20	Compliance Statement	>>If G18 contains NA then value equals NA; If G18 is equal to or less than G19, or G19 equals N/A, then report: Design Complies with the Total Allowed West-facing Fenestration Area; Else report: Total Proposed West-facing Fenestration Area Exceeds Allowable, Do Not Proceed >>
21	Proposed Fenestration U-factor (Windows)	>>If G02 = fixed window or operable window and if all associated values listed in column G09 are less than or equal to 0.30 then enter the single largest value from list; If G02 does not equal fixed window or operable window then value is NA; Else enter the weighted average value from the CF1R-ENV-02>>
22	Required Fenestration U-factor (Windows)	<<Report value = 0.30 F05a>>
23	Compliance Statement	>> If G21 equals NA then value equals NA; If G21 is equal to or less than G22 then report: Design Complies with the Maximum Allowed Fenestration U-value; Else report: Fenestration U-value Exceeds Allowable, Do Not Proceed
24	Proposed Fenestration SHGC (Windows)	<<If G02 = fixed window or operable window and if all associated values listed in column G11 or G14 are less than or equal to 0.23 then enter the single largest value from the two lists; Else If G11 equals NA, then report NA; If G02 does not equal fixed window or operable window then value is NA; Else enter the weighted average value from the CF1R-ENV-02>>
25	Required Fenestration SHGC (Windows)	<<If A09A07 = 1, 3, 5 or 16, then report NA; else report 0.23Report value = F06a>>
26	Compliance Statement	>> If G24 equals NA then value equals NA; If G24 is equal to or less than G25, or G25 equals N/A, then report: Design Complies with the Maximum Allowed Fenestration SHGC; Else report: Fenestration SHGC Exceeds Allowable, Do Not Proceed
27	Proposed Fenestration U-factor (Skylights)	<<If G02 contains skylight then enter the largest associated value from G09; If G02 does not contain skylight then value equals NA>>
28	Required Fenestration U-factor (Skylights)	<<If F01 contains 'Add Fenestration/Glazing ≤ 16 ft2 Skylight' then report 0.550.40; else report 0.30Report value = F05b>>
29	Compliance Statement	<< If G27 equals NA then value equals NA; If G27 is equal to or less than G28 then report: Design Complies with the Maximum Allowed Fenestration U-factor; Else report: Fenestration U-factor Exceeds Allowable, Do Not Proceed >>
30	Proposed Fenestration SHGC (Skylights)	<<_If G02 contains skylight then enter the single largest associated value from columns G11 or G14; Else If G11 equals NA, then report NA; If G02 does not contain skylight then value equals NA>>

31	Required Fenestration SHGC (Skylights)	<<If F01 contains 'Add Fenestration/Glazing ≤ 16 ft2 Skylight' then report 0.30; else report 0.23; Else if G11 or G14 equals NA, then report NAReport value = F06b>>
32	Compliance Statement	<< If G30 equals NA then value equals NA; If G30 is equal to or less than G31 Or If G31 equals NA then report: Design Complies with the Maximum Allowed Fenestration SHGC; Else report: Fenestration SHGC Exceeds Allowable, Do Not Proceed >>

H. Fenestration Proposed Areas and Efficiencies – Replace (Section 150.2(b)1B)

<<if A13 = Replace Fenestration/Glazing, Replace Fenestration/Glazing ≤ 75 ft2 window, or Replace Fenestration/Glazing Skylight then display this section; else display section header and the standard “This section does not apply” message>>

Note: <<Static Text - see the top>>

01	02	03	04	05	06	07	08	9	10	11	12	13	14
Tag/ ID	Fenestration Type	Frame Type	Dynamic Glazing	Orientation N, S, W, E	Area Removed (ft²)	Area Added (ft²)	Net Added Area (ft²)	Proposed U-factor	Proposed U-factor Source	Proposed SHGC	Proposed SHGC Source	Exterior Shading Device	Combined SHGC from CF1R-ENV-03
<<User input: Object NameP ermissi ve>>	<<User selects from list: *fixed window, *operable window, *Skylight>>	<<User selects from list: metal, metal thermal break, non- metal>>	<<User selects from list: None, Chromoge nic Glazing, Integrated Shading device>>	<<User selects from list: North, East, South and West>>	<<User input: Number> >	<<User input: Number >>	<<Calcula ted value: (H07- H06)>>	<<User input: DecimalN onnegativ e>>	<<User selects from list: NFRC, Tables 110.6-A and 110.6-B, Equations NA6-1 And NA6-2, or Area- weighted Average Workshee t (ENV- 02)>>-	<<If A09 = 1, 3, 5 or 16, then result equals NA; Else user input: DecimalN onnegativ e >>	<<User selects from list: NFRC, Tables 110.6-A and 110.6-B, Equations NA6-1 And NA6-2, or Area- weighted Average Worksheet (ENV-02) >>	<<User selects from list: Drop Arm Awnings, Louvered Sun Screen, Low Sun Angle Louvered Sun Screen, None, Operable Awnings, South- facing Overhang, Retractable Awnings, Roll Down Blinds Or Slats, Standard Bug Screen, Sun Screen, Vertical Roller Or Shades>>	<<If A09 = 1, 3, 5 or 16, then result equals NA; Elseif H13 equal to None, or H04 contains Chromogenic glazing, then report N/A; Else report value from CF1R-ENV-03 >>

15	Net Added West-facing Fenestration Area						<<Calculated value: sum of H08 where H05 = West>>						
16	Is Net Added Fenestration Area ≤ 0 for west-facing fenestration?						<<Calculated value: if H15 is less than or equal to 0 then value = Yes; else No>>						
17	Net Added Fenestration Area (all orientations)						<<Calculated value: sum of H08>>						
18	Is Net Added Fenestration Area ≤ 0 for all orientations?						<<Calculated value: if H17 is less than or equal to 0 then value = Yes; else No>>						
19	Proposed Fenestration U-factor (Windows)						<<If H02 does not contain fixed or operable window then report NA; If H02 contains fixed or operable window then enter the single largest associated value from H09>>						
20	Required Fenestration U-factor (Windows)						<<If F01 contains 'Replace Fenestration/Glazing ≤ 75 ft ² Window' then report 0.40; else report 0.30 Report value = F05a>>						
21	Compliance Statement						<<If H19 equals NA then value equals NA; If H19 is equal to or less than H20 then report: Design Complies with the Maximum Allowed Fenestration U-value; Else report: Fenestration U-value Exceeds Allowable, Do Not Proceed >>>>						
22	Proposed Fenestration SHGC (Windows)						<<If H02 does not contain fixed or operable window then report NA; If H02 contains fixed or operable window then enter the single largest associated value from columns H11 or H14; Else If H11 or H14 equals NA, then report NA>>						
23	Required Fenestration SHGC (Windows)						<<If A09A07 = 1, 3, 5 or 16, then report NA; else if F01 contains 'Replace Fenestration/Glazing ≤ 75 ft ² Window' then report 0.35; else report 0.23 Report value = F06a>>						
24	Compliance Statement						<<If H22 equals NA then value equals NA; If H22 is less than or equal to H23, or H23 equals NA, then report: Design Complies with the Maximum Allowed Fenestration SHGC; Else report: Fenestration SHGC Exceeds Allowable, Do Not Proceed >>>>						
25	Proposed Fenestration U-factor (Skylights)						<<If H02 does not contain skylight then report NA; If H02 contains skylight then enter the single largest associated value from H09>>						
26	Required Fenestration U-factor (Skylights)						<<Report value = 0.550-40F05b>>						
27	Compliance Statement						<<If H25 equals NA then value equals NA; If H25 is less than or equal to H26 then report: Design Complies with the Maximum Allowed Fenestration U-value; Else report: Fenestration U-value Exceeds Allowable, Do Not Proceed >>						
28	Proposed Fenestration SHGC (Skylights)						<<If H02 does not contain skylight then report NA; If H02 contains skylight then enter the single largest associated value from column H11 or H14; Else If H11 or H14 equals NA, then report NA>>						
29	Required Fenestration SHGC (Skylights)						<<Report value = 0.30; Else if H11 or H14 equals NA, then report NA F06b>>						
30	Compliance Statement						If H28 equals NA then value equals NA; If H28 is less than or equal to H29 Or If H29 equals NA then report: Design Complies with the Maximum Allowed Fenestration SHGC; Else report: Fenestration SHGC Exceeds Allowable, Do Not Report >>						

H1. Opaque Swinging Doors to Exterior (Section 150.1(c)5)

<<if A13 contains Opaque Exterior Doors, then display this section; else display the standard “This Section Does Not Apply” message>>

01	02	03	04	05	06	07
Tag/ID	Area	Proposed U-factor	Proposed U-factor Source	Required Maximum U-factor	Weighted Average (Yes/No)	Comments
<<user input: string (max 50 characters)>>	<<user input: DecimalNonnegative xxx.xx>>	<<user input: DecimalNonnegative x.xxx>>	<<user pick from list: *NFRC; or *Area-weighted Average Worksheet (ENV-02)>>	<<default value = 0.20>>	<<if H103 > 0.20, then value = Yes; Else value = No>>	<<user input: text>>

Note:

<<Static Text - see the top>>

H2. Space Conditioning (SC) Systems – Heating/Cooling – (Section 150.2(b))

<<

if none of the following are selected in A13: Space Heating System, or Space Cooling System, or Space Conditioning Duct System, then display the section does not apply message;

else if one or more of the following are selected in A13: Space Heating System, or Space Cooling System, or Space Conditioning Duct System, then require one row of data in this table and require one CF1R-ALT-02 and one CF2R-MCH-01b >>

01	02	03
Dwelling Unit Name	Dwelling Unit Total CFA (ft²)	Comments
<<User Input: ObjectNamePermissive>>	<<User Input: IntegerNonnegative>>	<<User Input: Text>>

J.K. Water Heating Systems (Section 150.2(b)1H)

List water heaters and boilers for both domestic hot water (DHW) heaters and hydronic space heating.

<<if Water Heating is not selected in A13, then display the section does not apply message; else display the section but entry is optional and allow user to add multiple rows>>

01	Is natural gas connected to the existing water heater? the existing water heater electric resistance?		<<User selects from yes or no>>			
02	03	0403	0504	06	0705	0806
Water Heating System ID or Name	Water Heating System Type	System Option (from §150.2(b)1Hiii)	Water Heater Type	Volume	Fuel Type	# of Water Heaters in System
<<user input text>>	<<user pick from list: *DHW;*Hydronic; *Combined Hydronic; >>	<<if JK01 = no, then user picks from list: 1, 2, 3, 4; Elseif JK01 = yes, then user picks from list: 1, 2, 3, 4; >>	<<if J043 = 1 then value = gas or propane water heater (Storage or instantaneous) or 4, then user select from Consumer instantaneous or Consumer Storage; if J04J03 = 2, then value = heat pump water heater; if JK043 is 3, then value = NEEA Tier 3 or higher heat pump water heater if K03 is 4, then value = Consumer electric water heater (Heat Pump Water Heater or Electric Resistance) >>	<<User input; if J05 = Consumer Instantaneous, value = NA>>	<< If JK043 = 1, then user picks from list *Natural gas, *Propane; elseif JK043 = 2 or 3, then value = Heat Pump; elseif JK043 = 4, then value = Electricity>>	<<user input: nonnegative number>>

Options:
<<Static Text - see the top>>